

4.11 Conditional expressions



This section has been set as **Optional** by your instructor. The activity will not show in s LMS. You can change settings when [configuring your zyBook](#).

Construct 4.11.1: Conditional expression.

```
expr_when_true if condition else expr_when_false
```

PARTICIPATION ACTIVITY

4.11.1: Conditional expression.

Start



2x speed

```
if condition:
    my_var = expr1
else:
    my_var = expr2
```

```
my_var = expr1 if (condition) else
```

Captions ^

1. This if-else form can be written as a conditional expression.
2. The condition in the middle is evaluated first.
3. If the condition evaluates to True, then expr1 is evaluated and my_var is assigned with expr
4. If the condition evaluates to False, then expr2 is evaluated and my_var is assigned with exp

**CHALLENGE
ACTIVITY**

4.11.1: Conditional expressions: Enter the output of the code.

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Start

Type the program's output

```
my_number = 7
your_number = 0 if my_number <= 9 else 4
print(your_number)
```

0

1

2

3

Check**Next**View solution ▼ *(Instructors only)***CHALLENGE
ACTIVITY**

4.11.2: Conditional expression: Print negative or positive.

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Drag and drop the code statement that assigns `cond_str` with a conditional expression that evaluates if `user_val` is less than 0, and "non-negative" otherwise. Then, the program outputs `user_val` followed by a space and `cond_str`.

Ex: If the input is `-9`, then the output is:

`-9 is negative.`

Note: Only one code statement on the left should be used in the final solution.

How to use this tool ▼

Unused

```
cond_str = "non-negative" if user_val < 0 else "negative"
"non-negative" if user_val < 0 else "negative"
cond_str = "negative" if user_val > 0 else "non-negative"
cond_str = "negative" if user_val < 0 else "non-negative"
```

main.py

```
user_val = int(input())
print(f"{user_val} is {cond_str}")
```

Check

View solution ▼ (Instructors only)

CHALLENGE ACTIVITY

4.11.3: Conditional expression: Conditional assignment.

Using a conditional expression, write a statement that increments `num_users` if `update_direction` is "up" and decrements `num_users` if `update_direction` is "down".

Sample output with inputs: `8 3`

`New value is: 9`

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```
1 num_users = int(input())
2 update_direction = int(input())
3
4 num_users = """ Your solution goes here """
5
6 print(f"New value is: {num_users}")
```

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